Machine Learning for Public Policy - Mini-Project 3

The University of Chicago - Harris School of Public Policy
PPHA 30545 - Professors Clapp and Levy
Fall 2025

This assignment must be handed in via Gradescope on Canvas by **11:45pm Central Time on Thursday, November 20th.**

You are welcome (and encouraged!) to form study groups (we recommend no more than 3 students) to work on the problem sets and mini-projects together. But you must write your own code and your own solutions. Please be sure to include the names of those in your group on your submission.

You should format your submission as a Jupyter Notebook (*.ipynb). Your notebook should include the code that produces your solutions and the accompanying output (e.g., estimates, tables, figures), as well as your answers and explanations written in Markdown.

Be sure to make it clear what question you are answering by labeling the sections of your write up well. Also, be sure that it is immediately obvious what supporting output from your code (e.g., estimates, tables, figures) you are referring to in your answers. In addition, your answers should be direct and concise. Points will be taken off for including extraneous information, even if part of your answer is correct. You may use bullet points if they are beneficial. Finally, for your code, please also be sure to practice the good coding practices you learned in *Data and Programming* and comment your code, cite any sources you consult, etc.

You are allowed to consult the textbook authors' website, Python documentation, and websites like StackOverflow for general coding questions. You may consult a large language model (LLM) or other AI tool (e.g., ChatGPT), but remember that you are only allowed to use AI tools in ways that are compliments to your learning, not substitutes for it (e.g., you should not ask ChatGPT to write large sections of your code or provide the answers to assignment questions). Please consult the AI Policy in the syllabus for more details, and ask if you are unsure. If you get help from an AI tool, you must provide in the query string you used and an explanation of how you used the AI tool's response as part of your answer. You are not allowed to consult material from other classes (e.g., old problem sets, exams, answer keys) or websites that post solutions to the textbook questions.

1 Overview

It is early in July of 2020, and you have just been hired to work for the Centers for Disease Control and Prevention (CDC).[1] As you probably know, the CDC is a national health protection agency tasked with protecting public health and safety by preventing and mitigating disease, injury, and disability in both the United States (US) and abroad. According to their website the CDC "conducts

[1] Your family is very proud and all of your friends are jealous of your great gig. You tell them you're so glad that you're going to take Machine Learning at Harris in the future, as it somehow really helped you land the job.

critical science and provides health information that protects our nation against expensive and dangerous health threats."

The number of COVID-19 deaths in the US has just surpassed 100,000. Things are looking grim around the world, but hope is on the way in the form of several potential vaccines. Your team at the CDC has been tasked with optimizing the distribution of a future vaccine to the places in the US that are most in need. To aid in doing so, you need a good prediction of which communities will be the most hard-hit when the vaccine is ready.

You colleagues have asked you to develop a model that predicts COVID-19 deaths per capita using socio-economic, health, and weather/pollution data from any and all available sources. Predictions based on your analysis will help shape your office's recommendations about how to best deploy limited vaccines. The project has two parts: understanding the available data and performing data analysis and answering questions.

2 Understanding the Data

The full dataset is available for download from Canvas, but you should familiarize yourself with how it was constructed. The data was compiled from the following sources:

1. NY Times: Daily data on COVID-19 cases and deaths at the county level comes from the New York Times GitHub Repository. Cumulative cases and deaths were calculated for counties in the continental United States as of June 1, 2020.[2]

2. Opportunity Insights: County characteristics, including baseline health measures, come from Bergeron, Chetty, Cutler, Scuderi, Stepner, and Turner (2016). The data are available for download on the Opportunity Insights Data Library webpage.[3]

3. PM COVID: Annual, county-level data on air pollution (PM2.5) and weather (average winter/summer temperature and relative humidity), and baseline mortality come from Wu, Nethery, Sabath, Braun, and Dominici (2020). The pollution and weather measures used in this project are calculated by averaging across annual averages from 2000-2016.[4]

3 Data Analysis[5]

1. The "Variable Description.xlsx" spreadsheet contains a list of variables that we'll use for our analyses. Note that this is not a full list of all the variables in the dataset, although it's

[2] See https://github.com/nytimes/covid-19-data for additional details. Note that three jurisdictions (New York, NY; Kansas City, MO; and Joplin, MO) report their counts at the city level instead of by county. In each of those instances, the cases/deaths were divided evenly among the overlapping counties. Note also that the CDC has similar data that can be disaggregated by demographic characteristics (gender, age group, race/ethnicity), but access to that data is restricted and requires securing a data-use agreement.

[3] See the description of the "County Characteristics (Described in eTable 9)" dataset for additional details.

[4] See https://github.com/wxwx1993/PM_COVID for additional details.

[5] Note that your responses in this section should be submitted and will be graded, but that some of the "questions" that follow explain how you should prepare the data and don't require that you submit anything.

close (we are ignoring a few perfectly co-linear predictors). Filter the full set of variables in the dataset down to the the Opportunity Insights and PM COVID variables listed in the spreadsheet along with `county`, `state`, and `deathspc`.[6]

2. Compute and report descriptive (summary) statistics for `deathspc` and the subset of Opportunity Insights and PM COVID variables you filtered in previous question.

3. Note that some variables have missing values. This causes problems when estimating the models. Normally we'd impute missing values, but to keep things simple, given the size of our data, you should drop all observations (rows) with missing values.

4. Create a separate dummy variable for each of the 48 states and the District of Columbia in the dataset (so you'll create 49 dummy variables in total, but dropping observations with missing values may reduce this number).

5. Split the sample into a training set (80%) and a test set (20%). Be sure to set a random state so you can replicate your work. There are odd patterns in the results with some splits of the data, so to avoid confusion and facilitate grading, please set `random_state=11`.

6. Using the training set, fit a model of COVID-19 deaths per capita ($y =$`deathspc`) as a function of the Opportunity Insights and PM COVID predictors listed in the spreadsheet, as well as state-level fixed effects (the state dummy variables) using OLS.

   (a) Using the model you fit, calculate and report the *MSE* in both the training and test sets.

   (b) Why might you be concerned about overfitting in this context? Is there any evidence of overfitting? Briefly explain.

7. Use the training set to estimate ridge regression and the lasso analogs to the OLS model in the previous question. For each, you should report a plot of the cross-validation estimates of the test error as a function of the value of the hyperparameter (λ) that indicates the tuned value of λ. Hint: to do so you should be sure to scale your predictors.[7] Then you should tune the hyperparameter by:

   (a) Calculating each model for a grid or range of values of λ. You will want to adjust the values you search over based on what you learn in the subsequent steps, but start by using 101 values of λ from 0.01 to 100.[8]

[6] Hint: rather than typing each variable name individually to select the variables to be summarized and for modeling, you can simply import them from the Excel file, add them to a list, and subset the data from that list. Also, note that the `county` variable is just an index to give real-world context to each observation. It will not be used for modeling.

[7] There are several ways to scale or normalize variables. Scaling here refers to transforming each predictor so that the mean is 0 and the variance is 1. Similar to the discussion in a previous assignment, there is a trade-off between scaling before splitting into training and test sets or after. The former uses more precise estimates of the population mean and variance. The latter does not bring information from the test set into the training set (which could result in inadvertent overfitting). Given the sample size in this question, you should scale *after* splitting your data.

[8] Hint: there are multiple ways to do so. Here's one: start by creating a range of 100 numbers from -2 to 2. You can do that by starting at -2 and incriminating each subsequent number by $\frac{1}{25}$. Let a denote those values and b denote the values of the grid for tuning λ. Calculate $b = 10^a$.

(b) Using 10-fold cross-validation (10FCV) (on the training data) to estimate the test error for each model at the given value of λ. Shuffle the data randomly for splitting. Please set `random_state=25` when doing so.

(c) Plotting the 10FCV estimates of the test error as a function of the value of λ.

(d) Choosing and reporting the optimal value of λ.

(e) Re-estimating your model on the full training set using that optimal value of λ.

8. Using the ridge regression and the lasso models you trained based on the optimal values of λ you found in the previous question, calculate and report the training and test set prediction errors (*MSE*) for each model. Did ridge regression and/or the lasso improve your prediction over OLS? Which model performs the best? Briefly explain which model you would recommend to the CDC and why.